

1 Gaussian Likelihood Derivation

1.1 Input and Assumption Nodes

Input node: `model.gaussian_observation`. Noise is iid $\mathcal{N}(0, \sigma^2)$.

1.2 Derivation

From $y = X\beta + \epsilon$, the residual vector $y - X\beta$ is Gaussian with covariance $\sigma^2 I_n$. Therefore

$$p(y \mid X, \beta, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \|y - X\beta\|_2^2\right).$$

1.3 Result

The result node is `likelihood.gaussian`.